DEFENSE INDUSTRIAL BASE SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military-Action Extension Methodology
Function-Primary Facility-Class Architecture — the Availability-and-Confidentiality Sector

DBA-DIB: Defense Industrial Base Sector Parent Document

A standalone sector series in the Design Basis for AI Governance in Critical Infrastructure

Version 0.5 — DRAFT

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For review by engineers, regulators, defense-industrial-base operators, and national-security and policy analysts responsible for the resilience and integrity of defense production, supply, and research.

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the National Infrastructure Protection Plan (NIPP) Defense Industrial Base sector. It is the parent instrument for the DBA-DIB series and applies the DB → DBA → DBA-MA three-layer analytical progression to the functions on which national defense-materiel capability depends: defense production and sustainment, the defense supply chain, and defense research, development, test and evaluation.

1.2 What This Document Does Not Do

This document does not supersede defense acquisition regulation, the classification system, or export-control law. It does not restate the Bright Line or Command Broker, inherited by reference from FD-BL. It does not re-scope the component production it depends on: the long-lead components and materials produced by the Critical Manufacturing sector are governed by DBA-CM and inherited here.

1.3 Function-Primary Organizing Principle

Placement follows the consequence chain. The sector is organized by function — the production and sustainment of defense materiel, the defense supply chain, and defense research and test — and each function is placed by the national-security consequence its failure or compromise produces, sub-indexed by security classification and sole-source criticality.

1.4 Series Placement — DBA-DIB as a Standalone Sector Series

DBA-DIB is a standalone sector series, a peer of the energy, water, chemical, communications, transportation, financial, healthcare, critical-manufacturing, emergency-services, and nuclear series. Its Sector Risk Management Agency is the Department of Defense. It inherits the DB → DBA → DBA-MA methodology from the DBA-MA parent and the Bright Line and Command-Broker architecture from FD-BL, and it applies the mandatory Military-Action Extension Applicability Determination.

1.5 Two Features No Other Sector Combines

The Defense Industrial Base is a producer sector — it manufactures weapons systems, aircraft, ships, and ammunition — but two features distinguish it from every other producer sector and shape its design basis. The first is that its governing threat is not only disruption but compromise. For every sector built so far the consequence has been an availability consequence: a facility fails and a service is lost until it recovers. For the Defense Industrial Base, the sector taxonomy records the defining condition — it is a prime target for nation-state espionage, and it handles classified information — and the consequence of compromise is a confidentiality-and-integrity consequence: the adversary does not merely deny the capability, it acquires the design or corrupts the supply chain. This is a consequence the corpus has not previously had to model, and

it is not recoverable: an exfiltrated weapon-system design and a pre-positioned supply-chain implant are not restored by rebuilding a plant.

The second feature is that the sector is downstream of Critical Manufacturing. The sector taxonomy names component production as a dependency, and the DBA-CM recovery-floor and concentration analysis — the sole-source, long-lead, foreign-ownership exposures — apply to the defense supply chain directly. The Defense Industrial Base therefore inherits the availability consequence of DBA-CM and adds to it the confidentiality-and-integrity consequence that is its own. The availability axis is measured, as in DBA-CM, by the recovery timeline for high-value long-lead components; the confidentiality-and-integrity axis requires a distinct, non-recoverable consequence category developed for the first time here.

Section 2 — The Defense Industrial Base and Its Regulatory Context

2.1 A Security-Governed, Sole-Source-Exposed Sector

The Defense Industrial Base is governed less by safety regulation than by security and acquisition regulation. The Department of Defense is the Sector Risk Management Agency and sets the terms through the Defense Federal Acquisition Regulation Supplement; the protection of controlled unclassified information is governed by NIST SP 800-171 and enforced through the Cybersecurity Maturity Model Certification; export of defense articles and technical data is governed by the International Traffic in Arms Regulations; and access to classified information is governed by the security-clearance system. The Defense Production Act, particularly Title III, is the mobilization lever. The sector's structural exposure is sole-source and limited-supplier concentration under a regime of long product-development cycles.

2.2 Regulator-and-Classification Inheritance by Function

The applicable regime varies by function: defense production inherits DFARS, the security-clearance framework, and the DPA mobilization authority; the defense supply chain inherits the CMMC and supply-chain-security regime and the DBA-CM component analysis; and research, development, and test inherit ITAR and the classified-information-handling requirements that make design exfiltration the governing consequence at that function.

2.3 The Regulatory Gap

The sector carries an extensive security and acquisition apparatus, but no instrument states a design basis in the corpus's consequence terms — the set of events, spanning both the availability and the confidentiality-and-integrity axes, that the defense-materiel base must be demonstrated to withstand. Security compliance is measured in controls and certifications; a design basis measured in demonstrated withstand-and-integrity capability does not exist.

Section 3 — Function-Primary Taxonomy

3.1 The Function Classes and the Classification / Sole-Source Sub-Index

The sector is decomposed into three function classes — defense production and sustainment, the defense supply chain, and research, development, test and evaluation — and one coordination class, the acquisition-and-mobilization coordination layer. Each is sub-indexed by security classification and sole-source criticality, so that the design basis load a given facility carries reflects both the sensitivity of what it holds and the irreplaceability of what it makes.

3.2 Defense Production & Sustainment (DP)

The defense-production function covers weapons-systems production, aerospace manufacturing, shipyards, and ammunition plants, together with the government-owned contractor-operated facilities, arsenals, and depots that produce and sustain materiel. Its governing consequence spans both axes: the availability loss of a sole-source production or sustainment capability, and the integrity compromise of a production line whose output enters the force. Its concentration exposure mirrors the long-lead-component exposure of DBA-CM in a defense-specific setting.

3.3 Defense Supply Chain (SC)

The defense-supply-chain function covers the sole-source component suppliers, the rare-earth materials, and the microelectronics fabrication on which defense production depends. It is the sector's tightest coupling to Critical Manufacturing, and it is the function where the confidentiality-and-integrity consequence is most acute: a supply-chain implant introduced at a component supplier propagates into every system that incorporates the component, undetected until activation. Its availability analysis is inherited from DBA-CM; its integrity analysis is developed here.

3.4 Research, Development, Test & Evaluation (RD)

The research-and-test function covers the research-and-development facilities, the testing ranges, and the test-and-evaluation systems where defense capability is designed and proven. Its governing consequence is the exfiltration of design and test data — the non-recoverable confidentiality consequence in its purest form — and it is the function at which the sector's classified-information-handling regime is most concentrated.

3.5 CorpCo — Acquisition & Industrial-Base Mobilization (coordination class)

The coordination function is carried by a coordination class spanning defense acquisition, the Defense Production Act Title III mobilization authority, and industrial-base policy. It is the locus at which national defense-production capacity is planned and mobilized, and the locus at which the AI-Governance Design Basis is implemented for the sector — the Bright Line placement and Command-Broker obligation for generative-AI and machine-learning systems in design,

manufacturing control, and acquisition, where the adversarial-integrity concern of Section 1.5 converges with the automation concern of the AI Governance design basis.

† The DP / SC / RD and CorpCo designators were ratified 2026-08-09.

3.6 The Dual-Axis Consequence

The sector requires two consequence measures, not one. On the availability axis it adopts the recovery timeline for high-value long-lead components (RT_HVLLC) inherited from the Critical Manufacturing and energy series, applied to sole-source defense production and supply. On the confidentiality-and-integrity axis it introduces a non-recoverable consequence category: the compromise is complete at the moment of exfiltration or implant and is not undone by recovery, so the design basis measure is the scope of capability compromised rather than a duration. Stating both axes explicitly, and refusing to collapse the second into the first, is the distinctive analytical requirement of this sector. *Defense Production and Sustainment* (DBA-DIB-DP-FC1) states both axes for the production function: it separates the confidentiality-and-integrity axis into an exfiltration chain and an integrity chain, applies RT_HVLLC as the production recovery time by way of *Long-Lead Critical Components* (DBA-CM-LL-FC1), and states the non-recoverable measure as the compromise scope of a data holding and the provenance coverage of delivered units (*Defense Production and Sustainment* §3.2, §3.3, §10.2); this framework takes the statement up at its next conformance pass.

Section 4 — The Defense Industrial Base Design Basis

[Skeleton — to be developed.] The design envelope will be organized by event category, spanning both the availability and the confidentiality-and-integrity axes, with a consequence-severity inclusion criterion consistent with the series.

4.1 Design Basis Principles

The criterion for inclusion is national-security consequence severity on either axis — potential for loss of a sole-source production, supply, or test capability, or for the compromise of a defense capability through exfiltration or supply-chain corruption — not event likelihood. Candidate bounding events, to be developed: loss of a sole-source production or sustainment capability; supply-chain compromise at a component or microelectronics supplier; exfiltration of design or test data from a research or test facility; and the compound kinetic-plus-cyber attack that denies a capability while compromising its replacement.

Section 5 — Dual Design Basis: Bright Line and Command Broker

[Skeleton — to be developed on the DBA-W §5.5 model.]

The Defense Industrial Base is both a heavy deployer of AI — in generative design, autonomous manufacturing, and acquisition analytics — and the sector where the adversarial-AI threat converges with the adversarial-integrity threat of Section 1.5. The Bright Line must be placed for

any action of a component within scope of FD-LD §4.3 whose incorrect or compromised performance could corrupt a design, a manufacturing process, or an acquisition decision at or above the severity threshold, with the architecturally enforced authorization of a qualified Command Broker. The inverter-firmware-trustworthiness precondition of the energy series generalizes here to a components-integrity precondition: a compromised or unverified AI or firmware component does not meet the determinism carve-out, whatever its apparent behavior, and each of its actions is placed on its own consequence (FD-LD §4.3; FD-BL §4.2). Where its output is an input on which placement or discharge depends and cannot be corroborated through an independent path, the action it bears on does not proceed without in-the-moment Command Broker authorization, and no placement changes (FD-BL-D2 §4.3). Supply-chain compromise is the sector's governing threat, which is why the precondition binds here. Determinism decides scope under FD-LD §4.3; it does not decide placement.

Section 6 — Cross-Sector Dependencies and the Consequence Elevation Rule

6.1 Inward View — Critical Manufacturing, Energy, Transportation, and IT

The sector's reverse dependencies are named directly in the sector taxonomy: Critical Manufacturing for component production — the defining upstream coupling, through which DBA-DIB inherits the DBA-CM recovery-floor and concentration analysis; energy for electricity and specialized fuels; transportation for the global supply chain and the strategic-mobility capacity established in the transportation framework; and information technology for secure communications and design systems. A further reverse dependency runs to the electric distribution sector through the defense-critical-load finding: the Defense Science Board's determination that the great majority of the most critical defense assets rely on commercial distribution infrastructure elevates those DisCo facilities under the cross-sector rule.

6.2 Outward View — National-Security Capability, and the Adversary's Gain

The outward consequence of the sector is national-security capability itself: degradation denies the ability to equip and sustain the force. But the sector's distinctive outward consequence is the one no other sector has — under the confidentiality-and-integrity axis, the adversary does not merely deny a capability, it acquires or corrupts one. The consequence flows outward as a gain to the adversary rather than only a loss to the defender, and the design basis must account for that asymmetry.

6.3 Boundaries with Critical Manufacturing and the Surveillance Standard

Two boundaries are named for clarity. The boundary with Critical Manufacturing runs at the point where a component enters a defense system: the general long-lead and concentration analysis is DBA-CM's, and the defense-specific integrity and classification analysis is DBA-DIB's. The boundary with the surveillance standard developed in the democratic-safeguards

work runs at the point where compromise shifts from materiel to influence: exfiltration and supply-chain corruption are DBA-DIB, while covert-influence operations against the industrial base fall under the surveillance sibling standard.

Section 7 — Military-Action Extension Applicability

[Skeleton.] The Defense Industrial Base is the sector where the military-action extension is most native: it is the base a military action is intended to defeat or deny. The persistence and denial characteristics dominate — pre-positioning within the supply chain, and the deliberate compromise of a capability before it is needed — and the exfiltration consequence is a military-action objective in itself. The sector-level presumption toward MA applicability is strong for sole-source production and for the microelectronics and design functions. Application remains a per-facility determination under the mandatory two-factor test — geolocation and strategic value — with strategic value driven by irreplaceability and by the sensitivity of what the facility holds.

Section 8 — Facility Class Register and Inheritance Model

8.1 Inheritance Model

Facility-class instruments of every type in the DBA-DIB series inherit the threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, and Bright Line / Command-Broker architecture established in this framework, and inherit the recovery-floor and concentration analysis of DBA-CM on their availability axis. Each instrument adds the defense-function-specific consequence chain on both the availability and the confidentiality-and-integrity axes.

8.2 Facility Class Register

The following table is the current facility-class register for the DBA-DIB series. DBA-DIB-DP-FC1 has issued as a working draft and the other function classes are at Anticipated status. Designators were ratified 2026-08-09 and are sub-indexed by security classification and sole-source criticality.

Designator	Function class (function-primary)	Governing characteristic	Status	Primary regulatory anchor
DBA-DIB-DP-FC1	Defense production & sustainment — weapons-systems production, aerospace, shipyards, ammunition plants, GOCO facilities, arsenals & depots	Materiel production/sustainment; sole-source concentration	Issued (working draft)	DoD (SRMA); DFARS; DPA Title III
DBA-DIB-SC-FC2	Defense supply chain — sole-source component suppliers, rare-earth materials, microelectronics fabrication	downstream of DBA-CM; supply-chain compromise	Anticipated	DoD; boundary with DBA-CM

Designator	Function class (function-primary)	Governing characteristic	Status	Primary regulatory anchor
DBA-DIB-RD-FC3	Research, development, test & evaluation — R&D facilities, testing ranges, T&E systems	design exfiltration; classified information	Anticipated	DoD; ITAR; NIST SP 800-171 / CMMC
DBA-DIB-CorpCo†	Defense acquisition & industrial-base mobilization (coordination class) — acquisition, DPA Title III, industrial-base policy	surge & mobilization coordination	Anticipated (architecture)	DoD; DPA Title III

Designators ratified 2026-08-09. Facility-class numbers are sequential across the series; CorpCo is a coordination class and carries no FC number, mirroring DBA-ES-CntlCo. Grounded in the CIKR_v16 sector map (Defense Industrial Base Core Infrastructure and Operational Characteristics).

Section 9 — Development Status and Open Items

This is the initial working draft of the DBA-DIB sector framework, grounded in the CIKR_v16 sector map. It establishes the standalone-series placement under the Department of Defense as Sector Risk Management Agency, the two distinguishing features (the confidentiality-and-integrity consequence axis alongside availability, and the downstream position relative to Critical Manufacturing), the function-primary decomposition (defense production, defense supply chain, research/development/test, and the CorpCo acquisition-and-mobilization class) with a classification-and-sole-source sub-index, the dual-axis consequence with a non-recoverable confidentiality category introduced for the first time, and the defense-critical-load and surveillance-standard boundaries. The design-basis envelope (Section 4), the dual-DB detail (Section 5), and the MA applicability section (Section 7) are stubs to be developed. Open items at this version: develop the non-recoverable confidentiality-and-integrity consequence category; and reconcile the DBA-CM component boundary and the surveillance-standard boundary. At v0.4, the series' first facility class has issued: DBA-DIB-DP-FC1 (*Defense Production and Sustainment*), physical, and the three reference events it proposed are conferred at Section 9.1. It develops, for the production function, the dual-axis consequence of Section 3.6 and the candidate bounding events of Section 4.1. What it notes to this framework is taken up at the framework's next conformance pass: the separation of the confidentiality-and-integrity axis into exfiltration and integrity chains, the six parameters and compromise scope as the non-recoverable measure, the sub-index stated on each axis separately, the event categories it develops ahead of Section 4, the Sustained Production and Bounded Compromise Conditions, its declared residuals and the consequence-basis conditionality check (*Defense Production and Sustainment* §10.9). The cover of v0.3 was dated August 2026 although that version was made in October 2026; v0.4 is dated October 2026. At v0.4 the Department of Defense is named Sector Risk Management Agency, the current name of the role, at Sections 1.4, 2.1 and 9, in the register and in the references (*Defense Production and Sustainment* §10.5). Further open item: the three reference-event instruments of Section 9.1. At v0.5, *Defense Production and Sustainment* at its v0.3 reads the Diehl fire under its §5.1 on the author's ruling of 14 October 2026 to follow the criminal record, which left its §5.2, sabotage of a production facility, without a case, and on the author's decision

of the same date it proposes a fourth event for that category, conferred at Section 9.1 as DBA-DIB-RE-4. Further open item: the instrument of DBA-DIB-RE-4.

9.1 Conferred Reference Events

On the author's decision of 14 October 2026, this framework confers a reference-event designator on each of the three events its first facility class proposed. A designator commits the series to two things: a row in the corpus-wide DBA Reference Event Register, and a standalone reference-event instrument for that event. Under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), reference-event instruments are written and held by Project 11, DBA Reference Events; this framework keeps the conferral. None of the three instruments is yet written. The designators are numbered in the order the proposing class lists its events; the number records conferral, not issue. Each designator is sector-primary and single.

From *Defense Production and Sustainment*, three events are conferred: DBA-DIB-RE-1, the fire at the Diehl group's plant in Berlin on 3 May 2024, whose attribution to sabotage is reported suspicion and unproven; DBA-DIB-RE-2, the campaign of Ukrainian strikes on Russian defense production, from 2024, of which the strike on the Alabuga plant on 23 April 2025 is the instance the class reads; and DBA-DIB-RE-3, the theft of data on the C-17 transport aircraft and on fighter aircraft from defense contractors between 2008 and 2014, on the Department of Justice's account. The Ukrainian event is a continuing campaign, and its instrument bounds it. The record of 155 mm production since 2022, which the class notes, is not conferred. The three instruments are owed, and they are Project 11's to write. *Defense Production and Sustainment* states the three as proposed at its v0.1 and conforms to the conferral at its next revision.

On the author's decision of 14 October 2026, a fourth event is conferred from *Defense Production and Sustainment* at its v0.3, for its §5.2, sabotage of a production facility: DBA-DIB-RE-4, the fire and explosions that destroyed the Canadian Car and Foundry Company's shell plant at Kingsland, New Jersey, on 11 January 1917, while the United States was neutral, on the record of the Mixed Claims Commission, United States and Germany, which found in 1930 that the fire was not caused by any German agent and in 1939, that finding set aside for fraud in the evidence Germany had presented, that Germany's liability was established. Its instrument is owed, and it is Project 11's to write; the class puts to it how the rule to follow the criminal record reaches a tribunal's award whose findings part (*Defense Production and Sustainment* §10.6). The three instruments of the first conferral were written by Project 11 on 14 October 2026, and DBA-DIB-RE-1 reads the Diehl fire under the class's §5.1, the criminal record having found no arson.

This framework is still a skeleton at Sections 4, 5 and 7. The three instruments read their events back through the design basis of *Defense Production and Sustainment*, which develops it ahead of the framework, and what they return to the framework is taken up at its next conformance pass.

References

[To be completed.] Governing and inherited instruments: FD-BL and determination annexes; DBA-MA Parent Framework (IAEA L1); DBA Series Gaps and Recommendations (§1.7, MA applicability); DBA-CM (component production; recovery-floor and concentration analysis; RT_HVLLC availability axis); DBA-TRAN (strategic-mobility supply chain); DBA-COMM / DBA-DC (secure communications and design systems); DBA-EN (defense-critical-load DisCo elevation, DSB finding); the surveillance sibling standard (Project 1, influence boundary). Regulatory: DoD (SRMA); DFARS; NIST SP 800-171; CMMC; ITAR; DPA Title III. Grounding: CIKR_v16 sector map (Defense Industrial Base Sector). Series instruments: DBA-DIB-DP-FC1 (*Defense Production and Sustainment*), the series' first facility class; DBA-DIB-RE-1 to DBA-DIB-RE-3, the conferred reference events of Section 9.1, their instruments owed to Project 11. DBA-DIB-RE-4, conferred at Section 9.1 at v0.5, its instrument owed to Project 11.

Revision History

0.1 August 2026 — T. Roxey. Initial skeleton, grounded in the CIKR_v16 sector map. Established DBA-DIB as a standalone sector series under DoD as SSA, distinguished by two features: a confidentiality-and-integrity consequence axis (espionage / exfiltration / supply-chain implant — a non-recoverable consequence category introduced for the first time) alongside the availability axis, and a downstream position relative to Critical Manufacturing (inheriting the DBA-CM recovery-floor and concentration analysis). Adopted the function-primary decomposition (Defense Production & Sustainment / Defense Supply Chain / Research, Development, Test & Evaluation / CorpCo) with a classification-and-sole-source sub-index; adopted the dual-axis consequence (RT_HVLLC availability + non-recoverable confidentiality/integrity); named the defense-critical-load (DSB) and surveillance-standard boundaries. Design-basis envelope, dual-DB detail, and MA applicability sections stubbed for development.

0.2 August 2026 — T. Roxey. Designators ratified; coordination class resolved to CorpCo (enterprise-internal; dominant DoD customer, DPA Title III active); DP / SC / RD confirmed; dual-axis consequence (availability + non-recoverable confidentiality/integrity) confirmed.

0.3 October 2026 — T. Roxey. Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026. Section 5 states the scope of Bright Line placement by the determinism carve-out of FD-LD §4.3 rather than as any AI action, and states that determinism decides scope and not placement (P9-11, wording). A compromised or unverified component is no longer placed below or above the line as a component: it does not meet the carve-out, its actions are placed on consequence, and an uncorroborated input on which placement or discharge depends takes in-the-moment Command Broker authorization with no change of placement (P9-10; FD-BL-D2 §4.3). Functional class is retired at every site under the author's determination of 30 September 2026, each site read for its sense: the decomposition tier reads function class, and the coordination layer reads coordination class (OI-15). No change to the design basis, the taxonomy or the register.

0.4 October 2026 — T. Roxey. Section 9.1 is added and confers DBA-DIB-RE-1 to DBA-DIB-RE-3 on the three reference events that DBA-DIB-DP-FC1 (Defense Production and

Sustainment) proposed, on the author's decision of 14 October 2026: the Diehl fire in Berlin (2024), the Ukrainian strikes on Russian defense production (from 2024) and the C-17 data theft (2008 to 2014). Each designator commits the series to a register row and a standalone instrument, none yet written; the instruments are Project 11's. The register at Section 8.2 enters the class as issued. Section 3.6 records the class's statement of both axes for the production function. Section 9 records the class's issue, what it notes to this framework, the correction of the cover date and the agency's name. The cover now reads October 2026; v0.3's cover read August 2026 although v0.3 was made in October. The Department of Defense is named Sector Risk Management Agency at Sections 1.4, 2.1 and 9, in the register and in the references. References add the class and the three designators. No change to the taxonomy, the designators of the classes, the sub-index or the design basis. Revision-history entries are left as records.

0.5 October 2026 — T. Roxey. Section 9.1 confers DBA-DIB-RE-4 on the event that DBA-DIB-DP-FC1 (Defense Production and Sustainment) at its v0.3 proposed for its §5.2, on the author's decision of 14 October 2026: the fire and explosions at the Canadian Car and Foundry Company's shell plant at Kingsland, New Jersey, on 11 January 1917, on the Mixed Claims Commission's record of 1930 and 1939. The designator commits the series to a register row and a standalone instrument, owed to Project 11. Section 9.1 also records that the first three instruments are written and that DBA-DIB-RE-1 is read under the class's §5.1. Section 9 and the references record the conferral. No change to the taxonomy, the designators of the classes, the sub-index or the design basis. Revision-history entries are left as records.